Tianzhi Xiang

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Research Interests

Acoustic Metasurfaces • **Wave Mechanics and Control** • **Acoustic–Elastic Wave Coupling**
Acoustic Communication • **Interfacial Wave Transmission**

Education

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2020 – Jun. 2024

- *B.Eng.* in Engineering Mechanics (National Strengthening Foundation Plan).
- **Core courses:** Theoretical Mechanics (94/100), Advanced Elasticity (96/100), Vibration and Measurement (90/100), Ordinary Differential Equations and Motion Stability (93/100).
- Undergraduate thesis: Optimal Design of Acoustic Metasurfaces with High Transmissivity Across Solid-Gas Interfaces (Supervised by Prof.Yanfeng Wang).

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2024 – Present

- *M.S.* Candidate in Mechanics | Expected Jun. 2027
- **Core courses**: Mechanical Metamaterials (99/100), Foundation of Solid Mechanics (93/100).

IMDEA Materials Institute, Madrid, Spain Aug. 2026 – Present

- Research Assistant supervised by Prof. Johan Christensen.

Publications & Patents

- **T. Z. Xiang**, H. T. Zhou, Y. Z. Tian, Y. F. Wang, J. Christensen, Y. S. Wang, "Vibro-acoustic metasurfaces for thickness-independent penetration of airborne sound through solid barriers." Under review at *International Journal of Mechanical Sciences*.
- H. T. Zhou, **T. Z. Xiang**, C. L. Han, Y. R. Jia, J. J. He, Y. F. Wang, Y. S. Wang, C. W. Qiu, "Reconfigurable Asymmetric Acoustic Transport Enabled by a Reciprocal Moiré Metasurface." Under review at *National Science Review*.
- **T. Z. Xiang** et al., "A Fabrication Method of an Acoustic Metasurface Structure for Efficient Sound Transmission Through Solid Barriers." *Chinese Invention Patent* CN119446109B, granted, 2025.

Research Experience

[Ongoing]Polarization Conversion by Vibro-Acoustic Metasurfaces Apr. 2026 – Present

Advisor: Prof. Yanfeng Wang & Prof. Johan Christensen

- Investigating angle-dependent polarization conversion between acoustic and elastic waves by metasurfaces.
- Analyzing scattering matrices and Fresnel coefficients to explore the relationship between polarization conversion and non-Hermitian phase transitions.
- Designing and optimizing asymmetric metasurfaces through numerical simulations, with experimental validation planned.

Vibro-acoustic metasurfaces for sound transmission enhancement through air-solid interfaces Sep. 2024 – Mar. 2026

Advisor: Prof. Yanfeng Wang

- Developed scattering matrix and interface matching models to explain the conversion of airborne sound into elastic waves in solids.
- Investigated how metasurfaces improve sound transmission across air-solid interfaces and reduce its dependence on barrier thickness.
- Demonstrated enhanced sound transmission through solid barriers using acoustic and vibration measurements.

Phase-Gradient Acoustic Metasurfaces for Wavefront Manipulation Jul. 2021 – Jun. 2023

Project Leader | Tianjin Municipal Undergraduate Innovation Training Program
Advisor: Prof. Yuesheng Wang & Prof. Yanfeng Wang

- Designed acoustic metasurfaces for wavefront control using Generalized Snell's Law and acoustic impedance models.
- Developed parametric unit cell models that provided a foundation for later research on vibro-acoustic coupling and interface matching.

PROJECTS

Reconfigurable Acoustic Moiré Metasurface Oct. 2024 – Jan. 2025

Role: **Second Author**.

- Conducted acoustic experiments to investigate tunable asymmetric sound transmission and backward suppression under different twist configurations.

RoboMaster University League Competition (DJI) Sep. 2022 – Jul. 2023

Role: **Team Leader**(Awarded National First Prize).

- Led a 40-member multidisciplinary team, coordinating project planning, task allocation, mechanical design reviews, and system integration throughout a one-year competition project.

2024 Global Youth Impact Hackathon (WFUNA) Dec. 2024

Role: Participant.

- Collaborated in a 48-hour multidisciplinary challenge to develop and present a plastic-pollution solution concept under SDG 14, covering feasibility and environmental impact.

HONORS & AWARDS

First-Class Academic Scholarship for Graduate Students	Sep. 2024
Outstanding Graduate of Tianjin University	Jun. 2024
Merit Student of Tianjin University	Dec. 2023
Outstanding Student Leader Award	Sep. 2021

TECHNICAL SKILLS

- **Language** English (IELTS Academic 7.0)
- **Software** COMSOL, SolidWorks, AutoCAD
- **Programming** MATLAB, Python, LaTeX
- **Computational Design** Genetic Algorithms, Topology Optimization
- **Experiments** Acoustic Measurement, Laser Vibrometry